

DRAWING LIST FOR TEST PACKAGE

AIR-SV-652

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	652	1FA	25A-AIR-SV-ST1-N-7022-01	1 OF 1	0	CJ-M15W13-PR-A20-000083		
2	652	1FA	25A-AIR-SV-ST1-N-7023-01	1 OF 1	0	CJ-M15W13-PR-A20-000083		
3	652	1FA	25A-AIR-SV-ST1-N-7024-01	1 OF 1	0	CJ-M15W13-PR-A20-000083		
4	652	1FA	50A-AIR-SV-ST1-N-7017-01	1 OF 1	0	CJ-M15W13-PR-A20-000083		
5	652	1MF	25A-AIR-SV-ST1-N-7021-01	1 OF 1	0	CJ-M15W13-PR-A20-000083		
6	652	1MF	40A-AIR-SV-ST1-C-7004-01	1 OF 4	0	CJ-M15W13-PR-A20-000083		
7	652	1MF	40A-AIR-SV-ST1-C-7004-02	2 OF 4	0	CJ-M15W13-PR-A20-000083		
8	652	1MF	40A-AIR-SV-ST1-C-7004-03	3 OF 4	0	CJ-M15W13-PR-A20-000083		
9	652	1MF	40A-AIR-SV-ST1-C-7004-04	4 OF 4	0	CJ-M15W13-PR-A20-000083		
10	652	1MF	40A-AIR-SV-ST1-C-7006-01	1 OF 4	0	CJ-M15W13-PR-A20-000083		
11	652	1MF	40A-AIR-SV-ST1-C-7006-02	2 OF 4	0	CJ-M15W13-PR-A20-000083		
12	652	1MF	40A-AIR-SV-ST1-C-7006-03	3 OF 4	0	CJ-M15W13-PR-A20-000083		
13	652	1MF	40A-AIR-SV-ST1-C-7006-04	4 OF 4	0	CJ-M15W13-PR-A20-000083		
14	652	1MF	50A-AIR-SV-ST1-C-7016-01	1 OF 1	0	CJ-M15W13-PR-A20-000083		
15	652	1MF	50A-AIR-SV-ST1-N-7017-01	1 OF 1	0	CJ-M15W13-PR-A20-000083		
16	652	2FA	15A-AIR-SV-ST1-N-7003-01	1 OF 1	0	CJ-M15W13-PR-A20-000083		
17	652	2FA	25A-AIR-SV-ST1-N-7018-01	1 OF 1	0	CJ-M15W13-PR-A20-000083		
18	652	2FA	25A-AIR-SV-ST1-N-7019-01	1 OF 1	0	CJ-M15W13-PR-A20-000083		
19	652	2FA	25A-AIR-SV-ST1-N-7020-01	1 OF 1	0	CJ-M15W13-PR-A20-000083		
20	652	2FA	40A-AIR-SV-ST1-N-7009-01	1 OF 1	0	CJ-M15W13-PR-A20-000083		

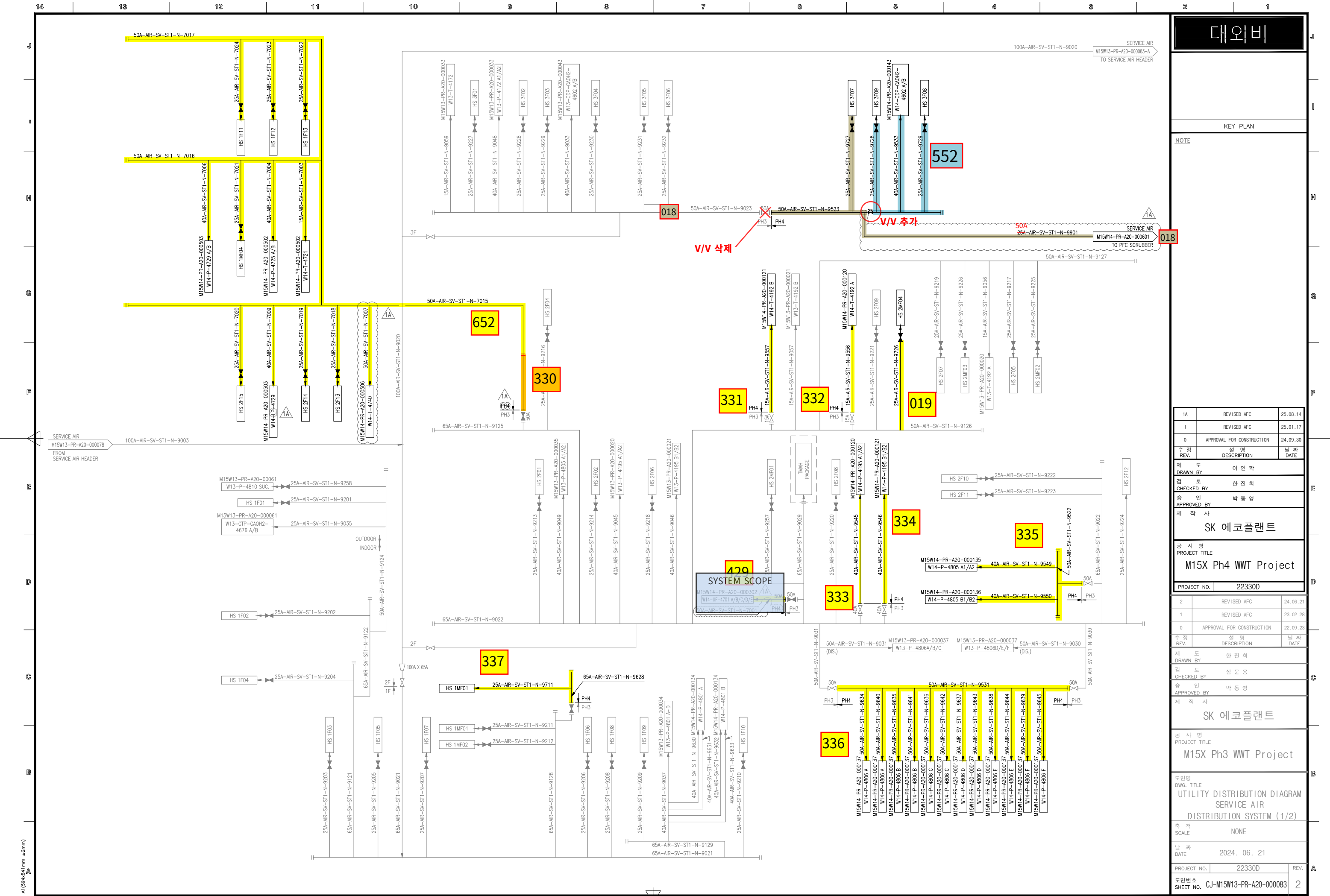
DRAWING LIST FOR TEST PACKAGE

AIR-SV-652

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
21	652	2FA	50A-AIR-SV-ST1-N-7007-01	1 OF 1	0	CJ-M15W13-PR-A20-000083		
22	652	2FA	50A-AIR-SV-ST1-N-7015-01	1 OF 2	0	CJ-M15W13-PR-A20-000083		
23	652	2FA	50A-AIR-SV-ST1-N-7015-02	2 OF 2	0	CJ-M15W13-PR-A20-000083		
24	652	2FA	50A-AIR-SV-ST1-N-7016-01	1 OF 1	0	CJ-M15W13-PR-A20-000083		

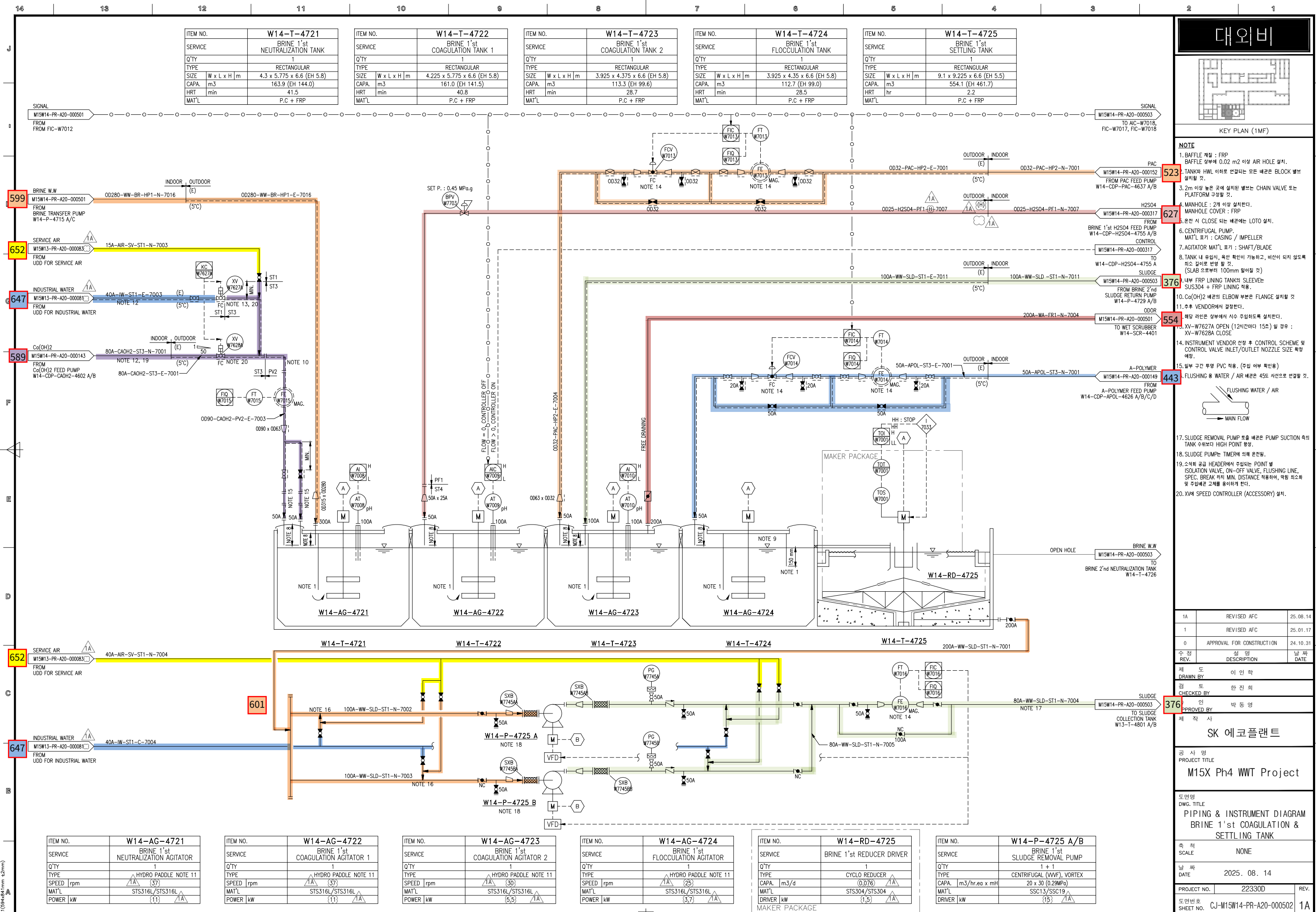


대외비

KEY PLAN

NOTE

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	설명 DESCRIPTION	날짜 DATE
제 도 DRAWN BY 이 인 학		
검 토 CHECKED BY 한 진 희		
승 인 APPROVED BY 박 동 영		
제 작 사 SK 에코플랜트		
공 사 명 PROJECT TITLE M15X Ph4 WWT Project		
PROJECT NO. 22330D		
2	REVISED AFC	24.06.21
1	REVISED AFC	23.02.28
0	APPROVAL FOR CONSTRUCTION	22.09.23
수정 REV.	설명 DESCRIPTION	날짜 DATE
제 도 DRAWN BY 한 진 희		
검 토 CHECKED BY 심 문 영		
승 인 APPROVED BY 박 동 영		
제 작 사 SK 에코플랜트		
공 사 명 PROJECT TITLE M15X Ph3 WWT Project		
도면명 DWG. TITLE UTILITY DISTRIBUTION DIAGRAM SERVICE AIR DISTRIBUTION SYSTEM (1/2)		
축척 SCALE NONE		
날짜 DATE 2024. 06. 21		
PROJECT NO. 22330D		REV.
도면번호 SHEET NO. CJ-M15W13-PR-A20-000083		2



대외비

KEY PLAN (1MF)

NOTE

1. BAFFLE 제형 : FRP
2. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
4. MANHOLE : 2개 이상 설치한다.
5. 온전 시 CLOSE 되는 배관에는 LOTO 설치.
6. CENTRIFUGAL PUMP.
7. AGITATOR MAT'L 표기 : SHAFT/BLADE
8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 반영 할 것.
9. (SLAB 으로부터 100mm 떨어진 것)
10. Co(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것
11. 주후 VENDOR에서 결정한다.
12. 해당 라인은 상부에서 시수 투입하도록 설치한다.
13. XV-W7627A OPEN (12시간마다 15초) 일 경우 : XV-W7628A CLOSE
14. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.
15. 일부 구간 투명 PVC 적용. (주입 여부 확인용)
16. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
17. SLUDGE REMOVAL PUMP 배관 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 형성.
18. SLUDGE PUMP는 TIMER에 의해 운전됨.
19. 소액의 공금 HEADER에서 주입되는 POINT 별 ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE, SPEC. BREAK 까지 MIN. DISTANCE 적용하여, 역행 최소화 및 주입배관 고체를 용이하게 한다.
20. XVM SPEED CONTROLLER (ACCESSORY) 설치.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31
수정	실명	날짜
REV.	DESCRIPTION	DATE
제 도	이 인 학	
DRAWN BY		
검 토	한 진 희	
CHECKED BY		
인	박 동 영	
APPROVED BY		
제 작 사		
SK 에코플랜트		
공 사 명		
PROJECT TITLE		
M15X Ph4 WWT Project		
도면명		
DWG. TITLE		
PIPING & INSTRUMENT DIAGRAM		
BRINE 1st COAGULATION & SETTLING TANK		
속 척		
SCALE		
NONE		
날 짜		
DATE		
2025. 08. 14		
PROJECT NO.		22330D
도면번호		CJ-M15W14-PR-A20-000502
SHEET NO.		1A

ITEM NO.	W14-T-4726
SERVICE	BRINE 2 nd NEUTRALIZATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 4.1 x 5.775 x 6.6 (EH 5.0)
CAPA.	m3 156.3 (EH 118.4)
HRT	min 34.1
MAT'L	P.C + FRP

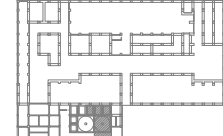
ITEM NO.	W14-T-4727
SERVICE	BRINE 2 nd COAGULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 4.2 x 5.775 x 6.6 (EH 5.0)
CAPA.	m3 160.1 (EH 121.3)
HRT	min 34.9
MAT'L	P.C + FRP

ITEM NO.	W14-T-4728
SERVICE	BRINE 2 nd FLOCCULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 4.7 x 5.775 x 6.6 (EH 5.0)
CAPA.	m3 179.1 (EH 135.7)
HRT	min 39.1
MAT'L	P.C + FRP

ITEM NO.	W14-T-4729
SERVICE	BRINE 2 nd SETTLING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 9.225 x 9.225 x 6.6 (EH 5.0)
CAPA.	m3 561.7 (EH 425.5)
HRT	hr 2.0
MAT'L	P.C + FRP

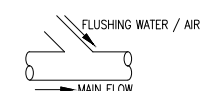
ITEM NO.	W14-LP-4729
SERVICE	LAMELLAR PLATE
Q'TY	1
CAPA.	W x L x H m BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 5,000m3/d)
MAT'L	ABS+SS304

대외비



KEY PLAN (1MF)

- NOTE
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m 이상 AIR HOLE 설치.
 2. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
 5. 온천 시 CLOSE 되는 배관은 LOTO 설치.
 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 반영 할 것.
(SLAB으로부터 100mm 떨어질 것)
 9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 10. 위에 것수등의 상세사상은 추후 VENDOR에서 결정한다.
 11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.
 12. FLUSHING 용 WATER / AIR 배관은 45도 서면으로 연결할 것.
 13. SLUDGE REMOVAL PUMP 또는 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 향상.
 14. DELETED.
 15. 추후 VENDOR에서 결정한다.
 16. DELETED.
 17. DELETED.
 18. DELETED.
 19. DELETED.
 20. SLUDGE PUMP는 TIMER에 의해 운전됨.



1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31

REV.	실명	날짜
1	이인학	
2	한진희	
3	박동명	

SK 에코플랜트

PROJECT TITLE

M15X Ph4 WWT Project

도면명

PIPING & INSTRUMENT DIAGRAM

BRINE 2nd COAGULATION & SETTLING TANK

SCALE NONE

날짜 2025. 08. 14

PROJECT NO. 22330D REV. 1A

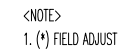
ITEM NO.	W14-AG-4726
SERVICE	BRINE 2 nd NEUTRALIZATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (37)
MAT'L	STS316L/STS316L
POWER	kw (1)

ITEM NO.	W14-AG-4727
SERVICE	BRINE 2 nd COAGULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (30)
MAT'L	STS316L/STS316L
POWER	kw (7.5)

ITEM NO.	W14-AG-4728
SERVICE	BRINE 2 nd FLOCCULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kw (3.7)

ITEM NO.	W14-RD-4729
SERVICE	BRINE 2 nd REDUCER DRIVER
Q'TY	1
TYPE	CYCLO REDUCER
CAPA.	m3/d (0.079)
MAT'L	STS304/STS304
POWER	kw (1.5)

ITEM NO.	W14-P-4729 A/B
SERVICE	BRINE 2 nd SLUDGE RETURN PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VVF), VORTEX
CAPA.	m3/hr.ea x mH 40 x 30 (0.29MPa)
MAT'L	SSC13/SSC19
POWER	kw (30)



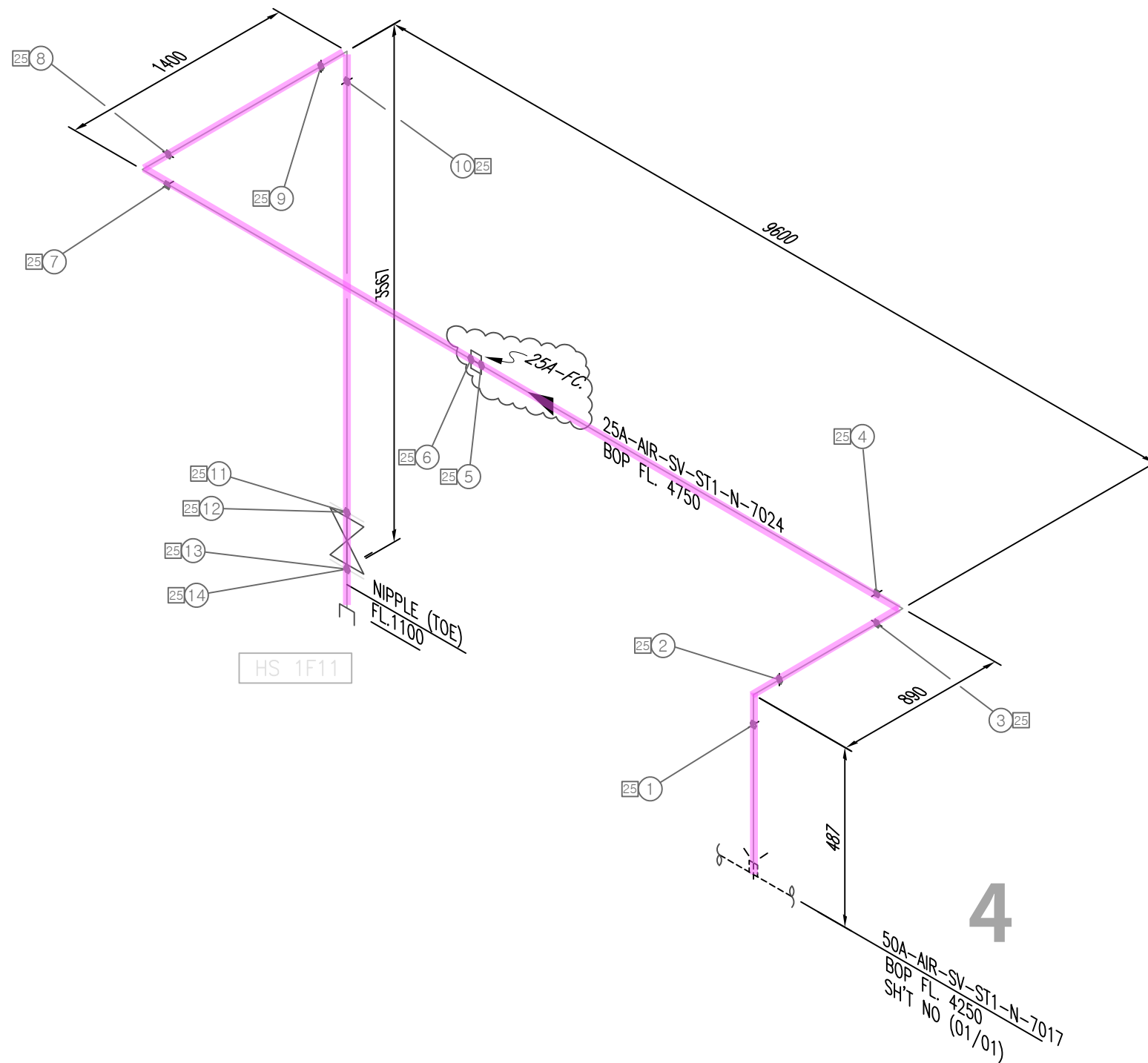
CLIENT		2	
			
CONTRACTOR		PROJECT NAME	
COMMODITY		JOB NO.	
AIR-SV		22330D	
LINE NO.	25A-AIR-SV-ST1-N-7023	SH'T NO.	REV.
		01 / 01	0

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CONFLD : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC/HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA : VG/VG1/VG2	* BALL VA : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA : VF/VF4	* CHECK VA : VC/NC1/NC2/NC3/NC4/NC5/NC6/NC7	* HOSE VA : VH/VH4	* PLUG VA : VP/VP1	* BALL VALVE : VB/VB1	* BALL VALVE : VB/VB1
* GLOBE VA : V/V1/V12	* DIAPHRAGM VA : VD/VD0	* DAMPER : DP	* QUICK COUPLING : SC	* FLEXIBLE JUNCTION(STEEL) : FLXJ	* FLEXIBLE JUNCTION(RUBBER) : FLXR	* FLEXIBLE JUNCTION(STEEL) : FLXJ	* FLEXIBLE JUNCTION(RUBBER) : FLXR	* FLEXIBLE JUNCTION(STEEL) : FLXJ	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SH	* SPECIAL BOLT : SB/BL1	
* STRAINER(TYPE) : YSTR	* STRAINER(TYPE) : YSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHIELD : SH	* SPECIAL BOLT : SB/BL1		

[illegible]



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	25A	15.9M	PIPE, STS 304TP, ERW, PE – SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE – SCH 40S
E90	25A	4EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
FC	25A	1EA	COUPLING FULL, STSF304, SW – SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLC'D, BODY–STSF304 /
			TRIM–STS304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLC SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)



<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT



3

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	13.5	60	UNIT	R1FA								M15X Ph4 Project		
			OPERATING	10	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-010016-B							COMMODITY	AIR-SV	JOB NO.	
						REF. P. & ID	CJ-M15W13-PR-A20-000083		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	25A-AIR-SV-ST1-N-7024	SH'T NO.	REV.
								REV. NO.	DATE	DESCRIPTION		CHK	APP'D			01 / 01	0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설, 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



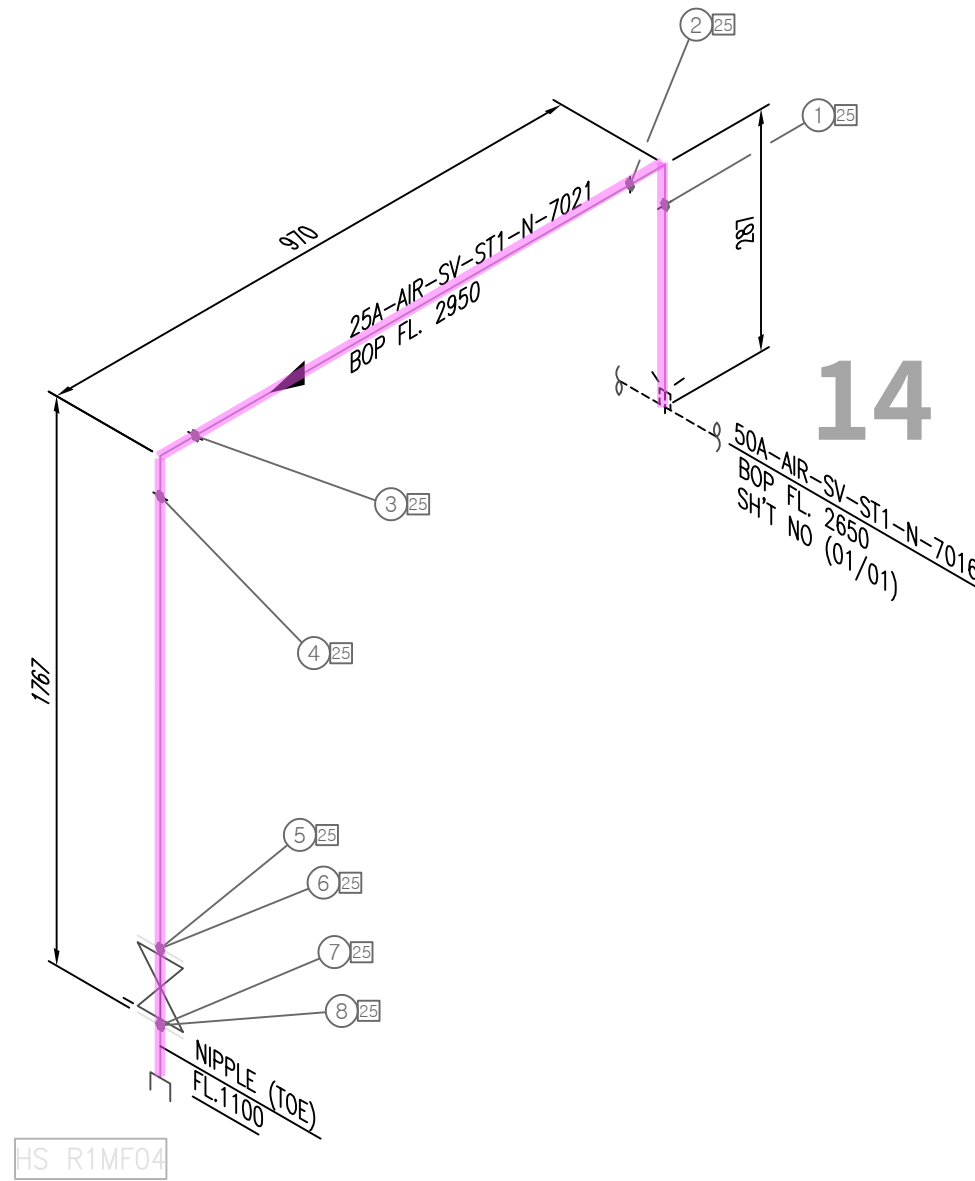
* COUPLING : FC * HALF COUPLING : HF/HF1
VC1/VC2/VC3/VC4/VC5/VC6/VC7
* EMERGENCY SHOWER : SSH
* SPECIAL BOLT & NUT : BL/BL1

CLIENT				4	
CONTRACTOR				PROJECT NAME M15X Ph4 Project	
COMMODITY AIR-SV		JOB NO. 22330D			
LINE NO. 50A-AIR-SV-ST1-N-7017		SH'T NO. 01 / 01		REV. 0	

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP (°C)	AREA	W14						
	THICK.	-	DESIGN	10	60	UNIT	R1FA						
			OPERATING	7.5	AMB	REF. PLAN DWG.							
						CJ-M15W14-PI-A30-010017-D			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
						REF. P & ID		REV.	DATE	DESCRIPTION	BY	CHK	APP'D
						CJ-M15W13-PR-A20-000081							



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	25A	3.0M	PIPE, STS 304TP, ERW, PE – SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE – SCH 40S
E90	25A	2EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY–STSF304 /
			TRIM–ST3304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)



<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA : VG/VG1/VG2	* BALL VA : VB/VB1/VB10/VB2/VB20	* BUTTERFLY VA : VF/VF4	* CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA : VL/VL1/VL2	* DIAPHRAGM VA : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT



5

[illegible]

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	40A	1.8M	PIPE, STS 304TP, ERW, PE – SCH 10S
E90	40A	2EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
VB	40A	1EA	BALL VALVE, KS 10K, RF FLC'D, BODY–STSF304 /
			TRIM–STS304 / SEAT–PTFE, FB, LEVER
F	40A	2EA	FLG SO, KS 10K, RF, STS304
G	40A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS



<NOTE>
1. (*) FIELD ADJUST

CLIENT



7

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	10	60	UNIT	R1MF								M15X Ph4 Project		
			OPERATING	7.5	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-010017-D		25.01.24	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	AIR-SV	JOB NO.	
						REF. P. & ID	CJ-M15W14-PR-A20-000502		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	40A-AIR-SV-ST1-N-7004	SH'T NO.	REV.
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			02 / 04	1

정고: 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유보자의 불법 사용으로 피해가 발생할 경우 유보자에게 민/형사상 책임을 요구할 수 있을

[illegible]

8

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB10/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	10	60	UNIT	R1MF								M15X Ph4 Project	
			OPERATING	7.5	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-010017-D		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	AIR-SV	
						REF. P. & ID.	CJ-M15W14-PR-A20-000502							LINE NO.	40A-AIR-SV-ST1-N-7004	
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D		03 / 04	0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음

[illegible]

9

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHIELDED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	10	60	UNIT	R1MF							M15X Ph4 Project		
			OPERATING	7.5	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-010017-D		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후		박동영	COMMODITY	AIR-SV
					REF. P. & ID	CJ-M15W14-PR-A20-000502	LINE NO.						40A-AIR-SV-ST11-N-7004		SH'T NO.	REV.
									REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D		04 / 04

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.

[illegible]

11

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VBLO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

[illegible]

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음

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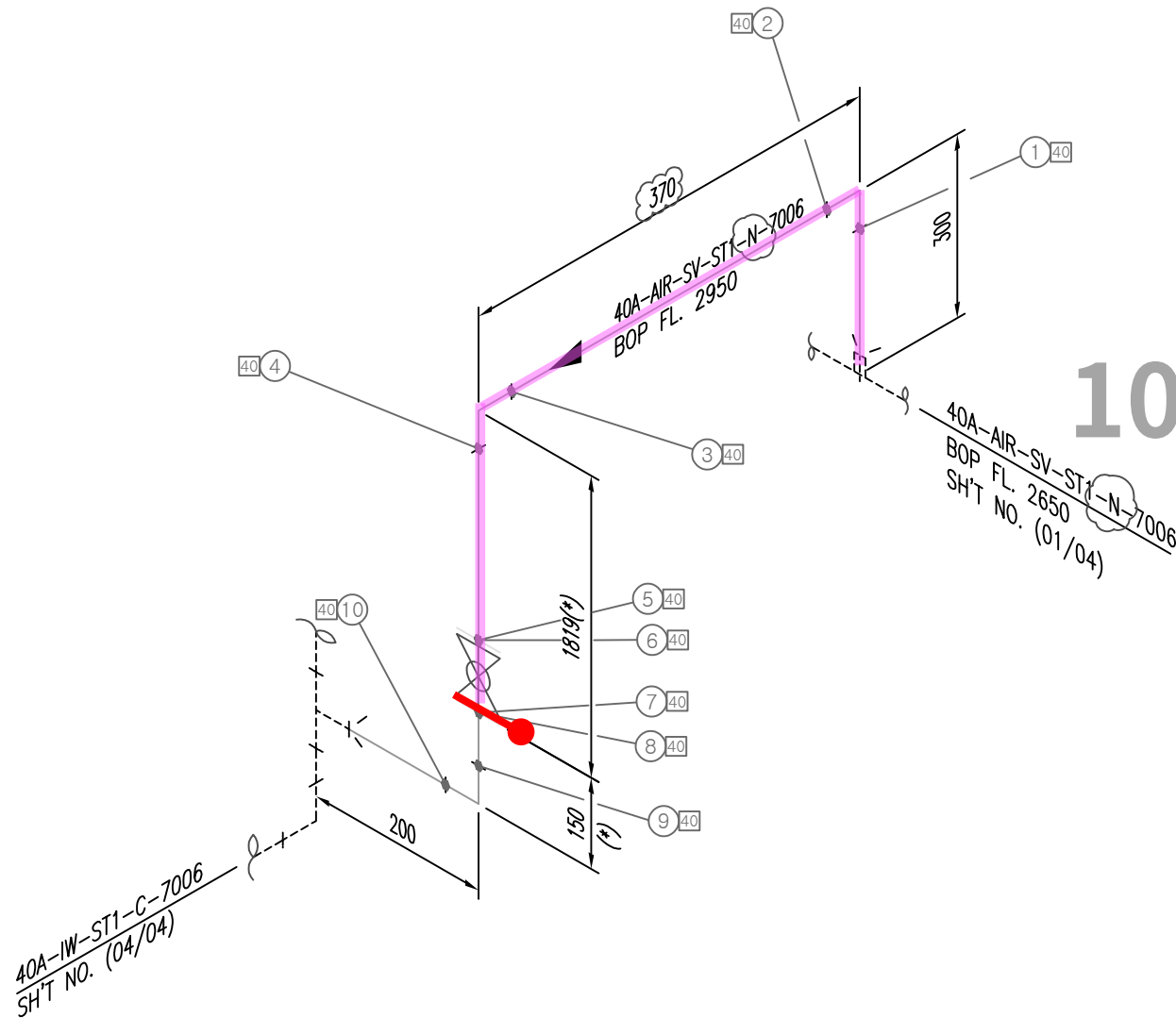
12

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VBLO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	10	60	UNIT	R1MF							M15X Ph4 Project		
			OPERATING	7.5	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-010017-D		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후		박동영	COMMODITY	AIR-SV
					REF. P. & ID	CJ-M15W14-PR-A20-000502	LINE NO.						40A-AIR-SV-ST1-N-7006		SH'T NO.	22330D
							REV. NO.						DATE		DESCRIPTION	BY

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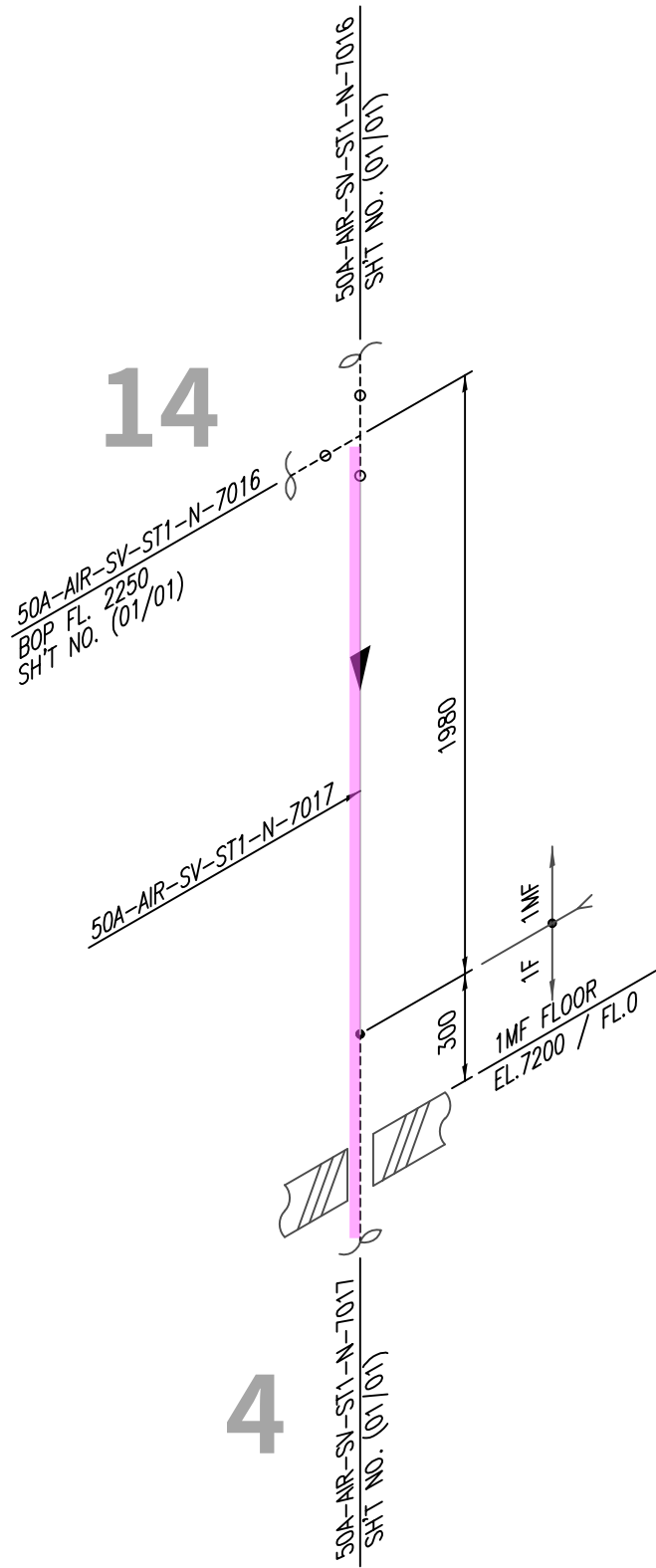
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		M15X Ph4 Project	
COMMODITY		JOB NO.	
AIR-SV		22330D	
LINE NO.		SH'T NO.	REV.
40A-AIR-SV-ST ¹ -N-7006		04 / 04	0



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA : VG/VG1/VG2	* BALL VA : VB/VB1/VB10/VB2/VB20	* BUTTERFLY VA : VF/VF4	* CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA : VL/VL1/VL2	* DIAPHRAGM VA : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	10	60	UNIT	R1MF								M15X Ph4 Project	
			OPERATING	7.5	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-010017-D		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	AIR-SV	JOB NO.
						REF. P. & ID	CJ-M15W14-PR-A20-000502							LINE NO.	40A-AIR-SV-ST1-N-7006	SH'T NO.



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME
	THICK.	—	DESIGN	10	60	UNIT	2FA										
			OPERATING	7.5	AMB	REF. PLAN DWG.			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			COMMODITY	JOB NO.
						REF. P. & ID											
						CJ-M15W14-PI-A30-010017-D		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			AIR-SV	22330D
						CJ-M15W13-PR-A20-000081											
																LINE NO.	SH'T NO.
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																	REV.
																	0

[illegible]

<NOTE>

1. (*) FIELD ADJUST

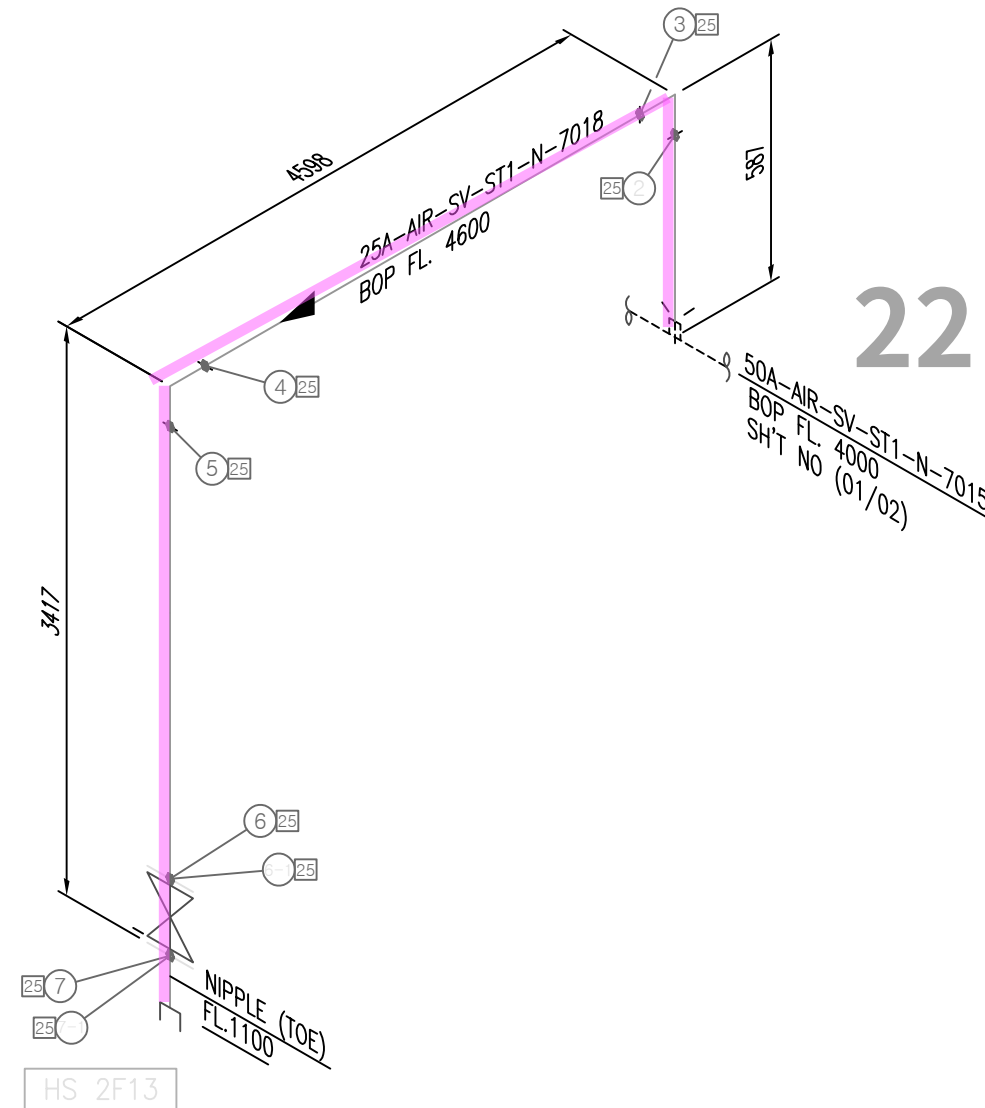
* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA : VG/VG1/VG2	* BALL VA : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA : VF/VF4	* CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA : VL/VL1/VL2	* DIAPHRAGM VA : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

[illegible]

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	25A	8.6M	PIPE, STS 304TP, ERW, PE – SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE – SCH 40S
E90	25A	2EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY–STSF304 /
			TRIM–ST3304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)



<NOTE>

1. (*) FIELD ADJUST

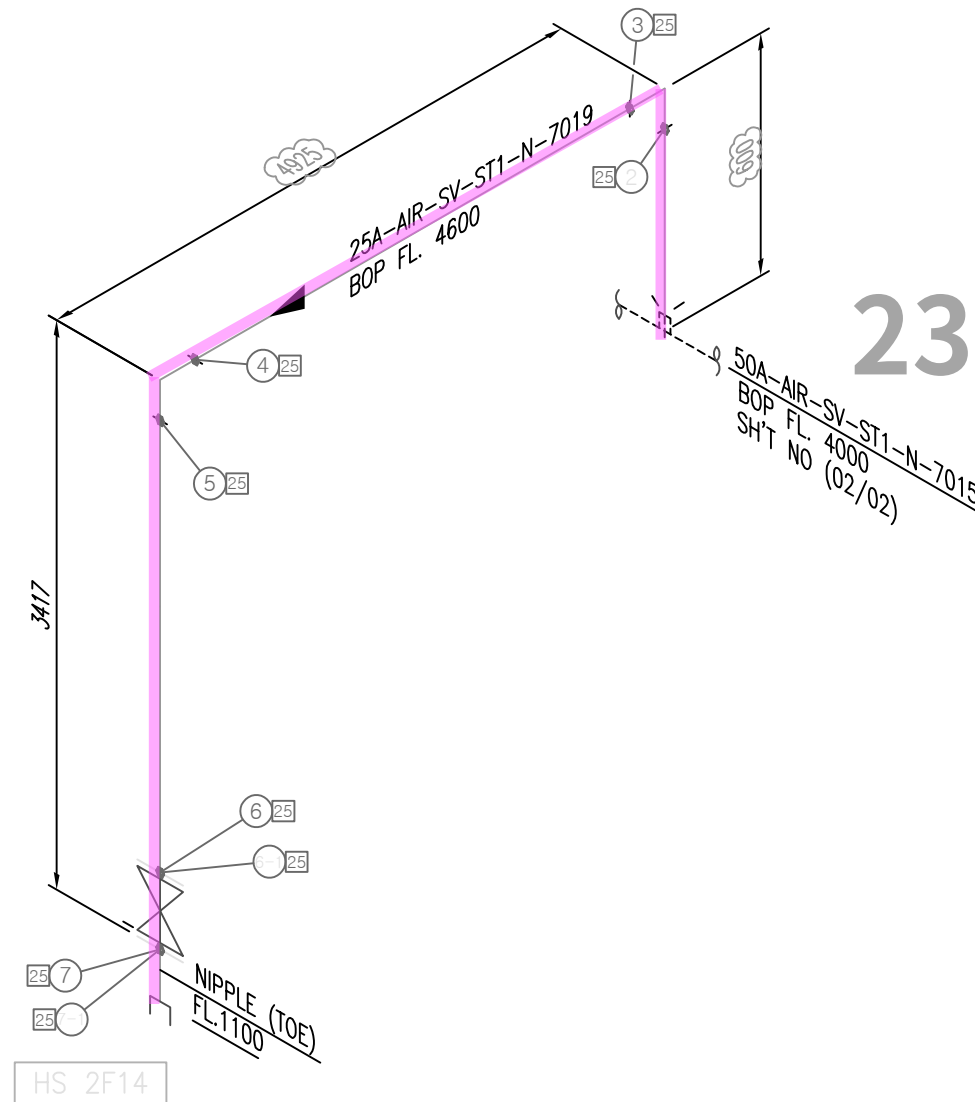
* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA : VG/VG1/VG2	* BALL VA : VB/VB1/VB10/VB2/VB20	* BUTTERFLY VA : VF/VF4	* CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA : VL/VL1/VL2	* DIAPHRAGM VA : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT		17	
			
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
AIR-SV		22330D	
LINE NO.		SH'T NO.	REV.
25A-AIR-SV-ST1-N-7018		01 / 01	0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	25A	8.4M	PIPE, STS 304TP, ERW, PE – SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE – SCH 40S
E90	25A	2EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY–STSF304 /
			TRIM–STSF304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)



<NOTE>

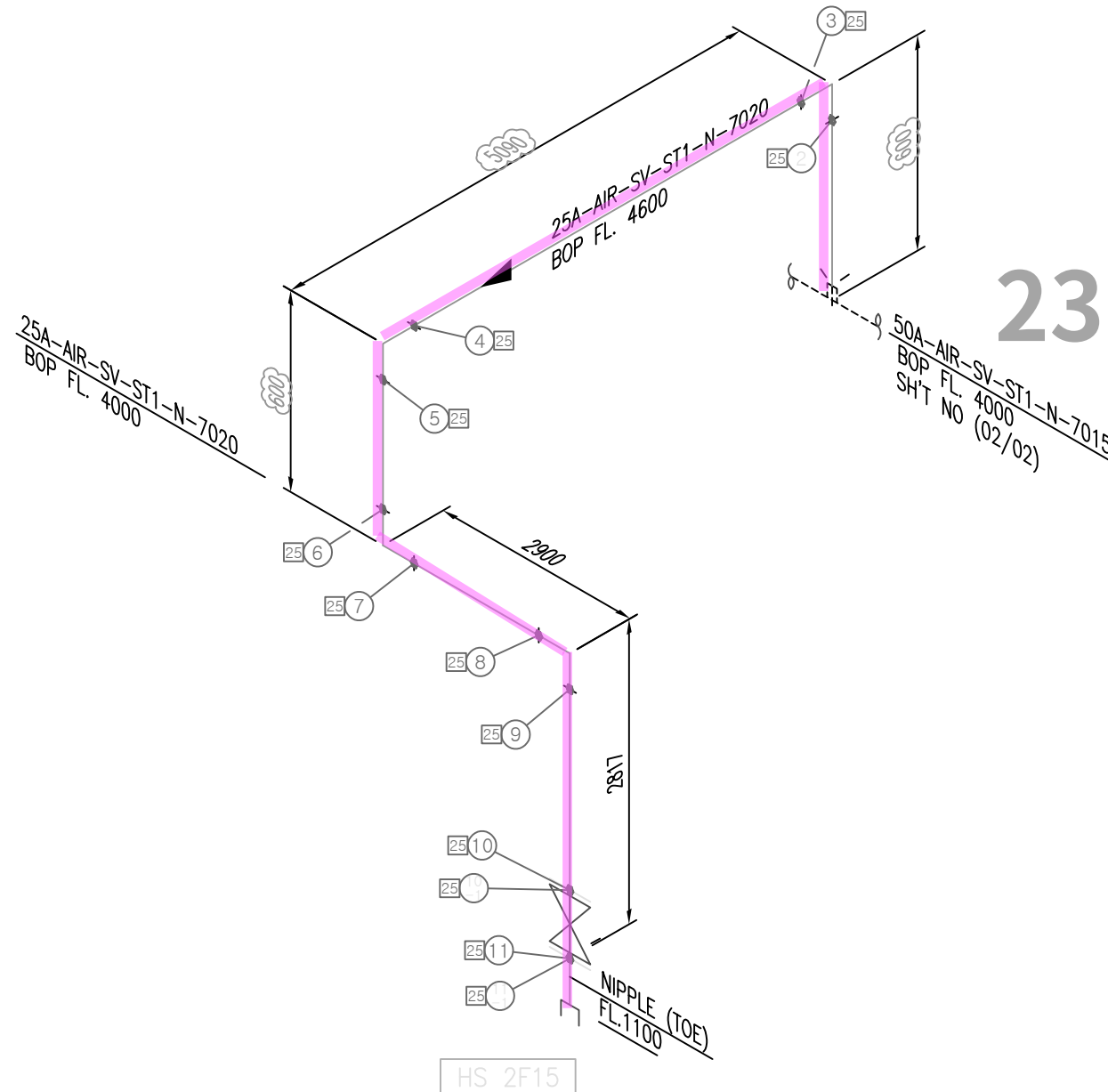
1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA : VG/VG1/VG2	* BALL VA : VB/VB1/VB10/VB2/VB20	* BUTTERFLY VA : VF/VF4	* CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA : VL/VL1/VL2	* DIAPHRAGM VA : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT		18	
		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
AIR-SV		22330D	
LINE NO.		SH'T NO.	REV.
25A-AIR-SV-ST1-N-7019		01 / 01	0



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	25A	11.5M	PIPE, STS 304TP, ERW, PE – SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, T0E – SCH 40S
E90	25A	4EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY–STSF304 /
			TRIM–STSF304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)



<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

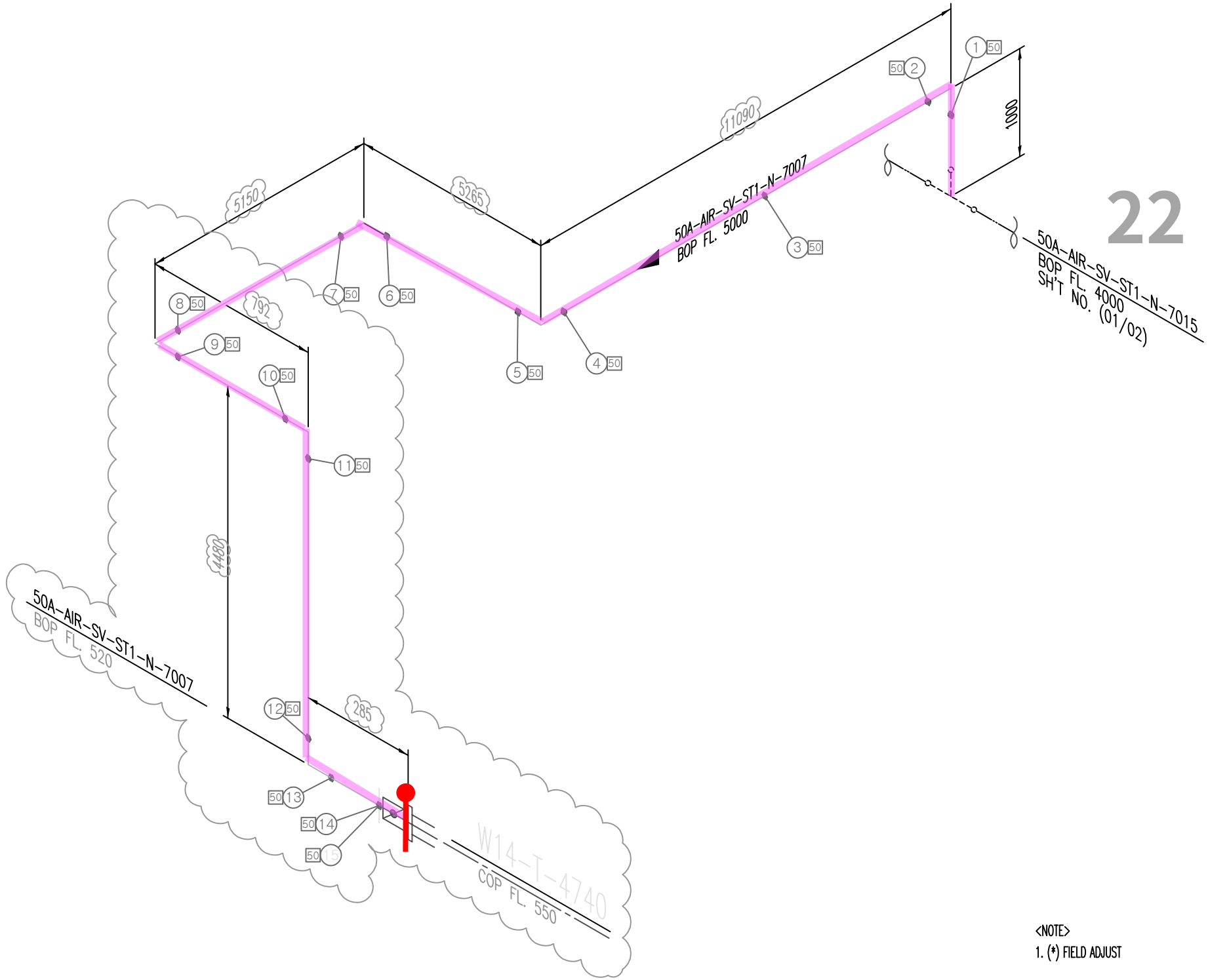
CLIENT



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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	13.5	60	UNIT	R2FA								M15X Ph4 Project		
			OPERATING	10	AMB	REF. PLAN DWG.							COMMODITY		AIR—SV		
						CJ—M15W14—PI—A30—020016—B		25.01.24	APPROVED FOR CONSTRUCTION	할래중	안남후	박동영		JOB NO.	22330D		
						CJ—M15W13—PR—A20—000083		REV. NO.	DATE	DESCRIPTION	B'D	CHK	APP'D	LINE NO.	25A—AIR—SV—ST1—N—7020	SH'T NO.	01 / 01

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	50A	26.4M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	50A	6EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
VF	50A	1EA	BUTTERFLY VALVE, KS 10K, WAFER RF, BODY-GCD450 / TRIM-ST304 / EPDM-LINED, LEVER
F	50A	1EA	FLG SO, KS 10K, RF, STS304
G	50A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
BL	50A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)

CLIENT



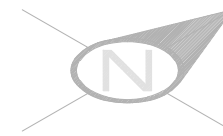
21

CONTRACTOR	PROJECT NAME
SK ecoplant	M15X Ph4 Project
COMMODITY	JOB NO.
AIR-SV	22330D
LINE NO.	SH'T NO. REV.
50A-AIR-SV-ST1-N-7007	01 / 01 1

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	10	60	UNIT	R2FA								M15X Ph4 Project		
			OPERATING	7.5	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-010017-D		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	AIR-SV	JOB NO.	
						REF. P. & ID	CJ-M15W13-PR-A20-000081		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	50A-AIR-SV-ST1-N-7007	SH'T NO.	REV.
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			01 / 01	1

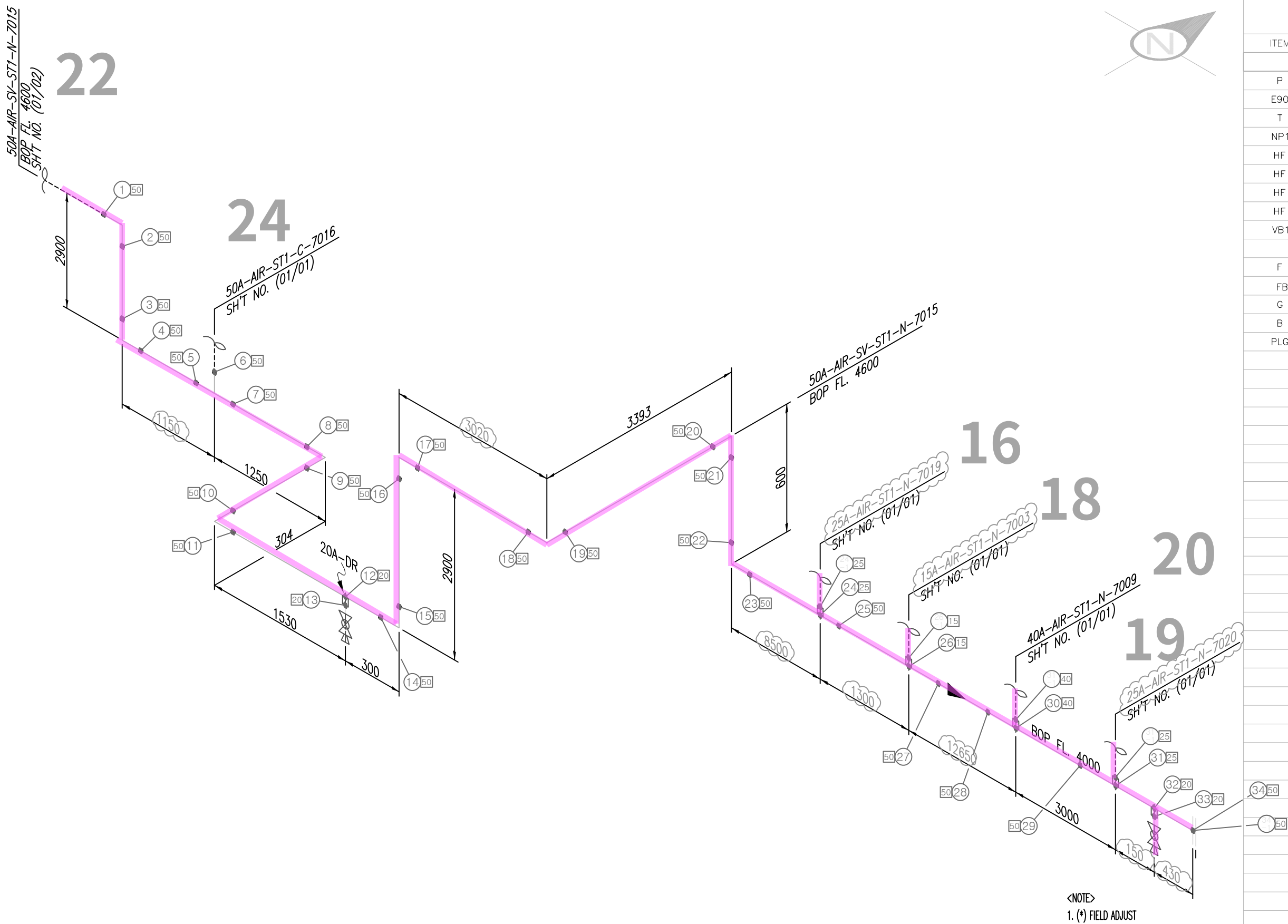
경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
AIR-SV		22330D	
LINE NO.		SH'T NO.	REV.
50A-AIR-SV-ST1-N-7015		01 / 02	0

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						
	THICK.	—	DESIGN	10	60	UNIT	R2FA						
			OPERATING	7.5	AMB	REF. PLAN DWG.							
						CJ-M15W14-PI-A30-010017-D		25.01.24	APPROVED FOR CONSTRUCTION	발령	안담 후	발령	
						REF. P. & ID CJ-M15W13-PR-A20-000081							REV. NO.

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	10	60	UNIT	R2FA								M15X Ph4 Project	
			OPERATING	7.5	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-010017-D		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	JOB NO.	
						REF. P & ID	CJ-M15W13-PR-A20-000081							LINE NO.	50A-AIR-SV-ST1-N-7015	
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D		02 / 02	0

[illegible]

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* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VBLO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	10	60	UNIT	R2FA							M15X Ph4 Project		
			OPERATING	7.5	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-020015-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후		박동영	COMMODITY	AIR-SV
					REF. P. & ID	CJ-M15W13-PR-A20-000081	LINE NO.						50A-AIR-SV-ST1-C-7016			
							REV. NO.						DATE		DESCRIPTION	BY